Parent Training for Dental Care in Underserved Children With Autism: A Randomized Controlled Trial

Rachel M. Fenning, PhD,[a,b] Eric M. Butter, PhD,[c] Eric A. Macklin, PhD,[d,e] Megan Norris, PhD,[c] Kimberly J. Hammersmith, DDS, MPH, MS,[f] Kelly McKinnon-Bermingham, MA, BCBA,[b] James Chan, MA,[d] Kevin G. Stephenson, PhD,[c] Charles Albright, PhD,[c] Jessica Scherr, PhD,[c] Jacquelyn M. Moffitt, BA,[a,b,g] Frances Lu, MPH,[d] Richard Spaulding, DDS,[h] John Guijon, DDS,[h,i] Amy Hess, BS,[c] Daniel L. Coury, MD,[c] Karen A. Kuhlthau, PhD,[j] Robin Steinberg-Epstein, MD[b]

abstract

OBJECTIVE: Children with autism spectrum disorder (ASD) have difficulty participating in dental care and experience significant unmet dental needs. We examined the efficacy of parent training (PT) for improving oral hygiene and oral health in underserved children with ASD.

METHOD: Families of Medicaid-eligible children with ASD (ages 3–13 years, 85% boys, 62% with intellectual disability) reporting difficulty with dental care participated in a 6-month randomized controlled trial comparing PT ($n = 60$) with a psychoeducational dental toolkit ($n = 59$). Primary outcomes were parent-reported frequency of twice-daily toothbrushing and dentist-rated visible plaque. Secondary outcomes included parent-reported child behavior problems during home oral hygiene and dentist-rated caries. Dentists were blind to intervention assignment. Analyses were intention to treat.

RESULTS: Retention was high at posttreatment (3 months, 93%) and 6-month follow-up (90%). Compared with the toolkit intervention, PT was associated with increased twice-daily toothbrushing at 3 (78% vs 55%, respectively; $P < .001$) and 6 (78% vs 62%; $P = .002$) months and a reduction in plaque at 3 months (intervention effect, −0.19; 95% confidence interval [CI], −0.36 to −0.02; $P = .03$) and child problem behaviors at 3 (−0.90; 95% CI, −1.52 to −0.28; $P = .005$) and 6 (−0.77; 95% CI, −1.39 to −0.14; $P = .02$) months. Comparatively fewer caries developed in children receiving the PT intervention over 3 months (ratio of rate ratios, 0.73; 95% CI, 0.54 to 0.99; $P = .04$).

CONCLUSIONS: PT represents a promising approach for improving oral hygiene and oral health in underserved children with ASD at risk for dental problems.

Full article can be found online at www.pediatrics.org/cgi/doi/10.1542/peds.2021-050691

[a]*Department of Child and Adolescent Studies and Center for Autism, California State University, Fullerton, Fullerton, California;* [b]*The Center for Autism and Neurodevelopmental Disorders, Department of Pediatrics, School of Medicine, University of California, Irvine, Irvine, California;* [c]*Departments of Pediatrics and* [f]*Dentistry, Nationwide Children's Hospital, Columbus, Ohio;* [d]*Biostatistics Center, Department of Medicine, Massachusetts General Hospital, Boston, Massachusetts;* [e]*Department of Medicine, Harvard Medical School, Boston, Massachusetts;* [g]*Department of Psychology, University of Miami, Coral Gables, Florida;* [h]*Healthy Smiles for Kids of Orange County, Garden Grove, California;* [i]*Pediatric Dentistry, Ostrow School of Dentistry, University of Southern California, Los Angeles California; and* [j]*Department of Pediatrics, Massachusetts General Hospitaland Harvard Medical School, Boston, Massachusetts*

Drs Fenning and Butter conceptualized and designed the study, developed the treatment curriculum and data collection instruments, supervised data collection, drafted the initial manuscript, and critically reviewed and revised the manuscript; Dr Macklin conceptualized and designed the study, developed data collection instruments, supervised data collection, performed analyses, drafted the initial manuscript, and critically reviewed and revised the manuscript; Drs Norris, Kuhlthau, and Steinberg-Epstein conceptualized and designed the study, developed data collection instruments, supervised data collection, drafted the initial manuscript, and critically reviewed and revised the manuscript; Dr Coury and Ms Hess

WHAT'S KNOWN ON THIS SUBJECT: Children with autism spectrum disorder have difficulty participating in dental care and experience significant unmet dental needs, which may be exacerbated by underserved status. No known interventions have improved oral hygiene or oral health in this population in randomized trials.

WHAT THIS STUDY ADDS: In a randomized trial involving underserved children with autism spectrum disorder and dental care problems, parent training significantly improved oral hygiene and health relative to a psychoeducational toolkit. Parent training represents a promising oral health intervention for this high-risk population.

To cite: Fenning RM, Butter EM, Macklin EA, et al. Parent Training for Dental Care in Underserved Children With Autism: A Randomized Controlled Trial. *Pediatrics*. 2021;149(5):e2021050691

One of the most common unmet child health care needs, dental care presents a particular challenge for children with autism spectrum disorder (ASD),[1] a population with an elevated risk for plaque, caries, and oral health problems across numerous studies,[2–7] although not universally.[8,9] Children with ASD exhibit greater distress during dental care and experience increased difficulty with participating in routine oral hygiene and dental visits than children without ASD.[10–12] Challenges with participating in dental care may compromise care quality[13,14] and may adversely influence parental perspectives about future dental care.[11] Multiple factors may impede dental care for children with ASD, including child ASD symptoms, comorbid intellectual disability (ID), communication difficulties, and challenging behaviors.[1,15–17] Barriers associated with low income and racial or ethnic minority status may compound difficulties in obtaining dental care.[1,15]

Dental interventions for children with ASD have predominantly focused on improving compliance with dental office procedures.[18–33] Less attention has been devoted to oral hygiene, despite evidence of poor oral hygiene in children with ASD[6,10,12,34,35] and close links among oral hygiene, oral health, and overall well-being.[36–38] Leveraging work with parents to promote oral hygiene in children with ASD holds promise given the fundamental role parents play in supporting oral hygiene and oral health for all children[36,38–41] and the heightened influence of parental support for children with ASD,[42] particularly among children with comorbid ID.[43]

Daily living skills have long been targeted through behavioral intervention,[44] but few studies have focused specifically on oral hygiene in children with ASD. In a multiple-baseline study, investigators used gradual exposure to increase and generalize toothbrushing compliance in 3 children with ASD.[45] Parent-implemented visual supports have been associated with increased toothbrushing participation[46–48] and improved oral health in children with ASD in pre- and postintervention designs,[48–50] and researchers of a randomized trial found promising, but nonsignificant effects of video modeling on oral health.[51] Additionally, in an uncontrolled trial, researchers linked caregiver psychoeducation and training in toothbrushing to improved oral health in individuals with ASD.[52] Collectively, findings reveal that parent-implemented intervention may be a useful approach for improving oral hygiene and oral health in children with ASD. However, randomized controlled trials (RCTs) are lacking, as are interventions that target both oral hygiene behavior and oral health outcomes. Additional attention to economically disadvantaged families is also needed.

In the current investigation, we have addressed these limitations by developing and testing a parent training (PT) intervention designed to improve primary outcomes of daily home oral hygiene and oral health in underserved families of children with ASD. In this RCT, we examined the efficacy of PT (R. Fenning, K. McKinnon-Bermingham, and E. Butter, unpublished manual, 2017) relative to a psychoeducational dental toolkit for increasing frequency of twice-daily home toothbrushing and reducing visible plaque. We expected that PT would significantly improve oral hygiene and oral health relative to the toolkit at postintervention (3 months, secondary end point) and that relative gains would be maintained at 6-month follow-up (primary end point). Secondary aims examined the efficacy of PT for reducing severity of child behavior problems during home dental care and in preventing caries. Post hoc exploratory analyses investigated changes in oral hygiene as mechanisms underlying PT effects on oral health.

METHODS

Participants

Families of children with ASD and Medicaid eligibility were recruited from December 2016 to December 2018 for participation in a 6-month RCT at 2 hospital-affiliated sites in a large research network. Follow-up concluded in June 2019. Additional eligibility criteria included parent-reported difficulty participating in dental care, child age 3 to 13 years at enrollment, child ASD diagnosis confirmed through clinical best estimate on the basis of the *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition*,[53] research-reliable administration of the Autism Diagnostic Observation Schedule-Second Edition,[54] and parent-reported absence of child dental screenings and examinations in the previous 6 months. Exclusion criteria included an acute dental condition requiring immediate emergency treatment, medication-induced oral health adverse effects, and participation in nonstudy dental interventions. Study procedures were conducted in English.

Phone screening was completed with 266 families. Of these, 157 families consented to baseline assessments, and 119 were eligible for study participation. Eligible participants were randomly assigned by the data coordinating center using a concealed computer-generated, site-stratified, permuted block schedule that assigned participants equally to the PT ($n = 60$) or toolkit ($n = 59$) intervention. Our 3 study dentists

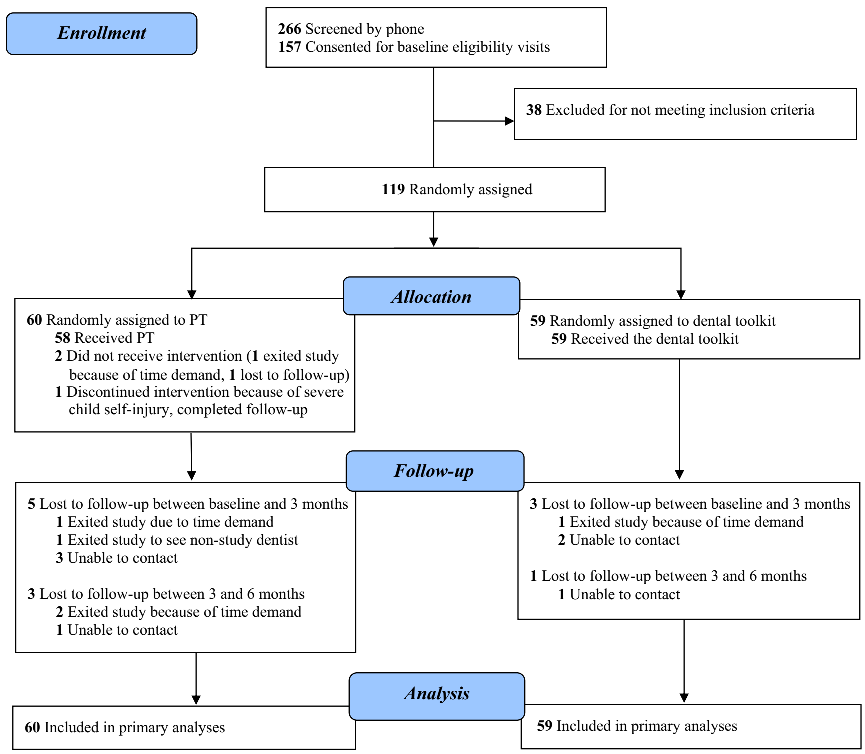

FIGURE 1
Flow of families through an RCT of PT versus a psychoeducational dental toolkit.

were blinded to intervention assignment. Retention was high (90%), with 107 families completing the final follow-up (Fig 1).

Procedures

An independent data and safety monitoring board oversaw study procedures, which were approved by site institutional review boards (2016P001195/PHS, HS-2016-3043, IRB16-00736, HSR-17-18-580). Parents provided written informed consent, and children provided assent when appropriate. Phone screening was used to determine demographic eligibility. Prerandomization baseline assessments included parent questionnaires, a dental visit that yielded standardized ratings of oral health, and a clinic visit that involved testing of child intellectual functioning and ASD symptoms. Follow-up dental visits and parent questionnaires provided outcome measures at 3 and 6 months after baseline.

Participants were seen by the same board-certified pediatric dentist at each time point. Reliability for oral health measures was established prestudy using photographs; during the trial, dentists engaged in ongoing calibration through monthly calls and annual in-person exercises. Dental visits involved a standard chair, overhead light, visual examination of full-mouth caries followed by plaque ratings on index teeth, and preventive cleaning, adjusted for children's presenting needs. Given our population, dental charting did not include a dental explorer or x-rays.

Dental Toolkit

The toolkit was an active, multicomponent comparator involving (1) the Autism Speaks Autism Treatment Network/Autism Intervention Research Network on Physical Health Dental Toolkit, a family-friendly psychoeducational resource about oral hygiene and dental visits; (2) an Oral-B Vitality Floss Action electric toothbrush to support plaque reduction[55]; (3) a 6-month supply of oral hygiene materials (toothbrush heads, fluoride toothpaste, and flossers), and (4) mock dental tools (eg, dental mirror). Manual toothbrushes were provided to encourage family-wide oral hygiene.

PT to Improve Dental Care

Families participating in PT received all toolkit components and the novel intervention curriculum (R. Fenning, K. McKinnon-Bermingham, E. Butter, unpublished manual, 2017) outlined in Table 1. Individual PT comprised 7 core in-person sessions, including a home visit and a dental office coach, and 4 phone booster sessions. PT included psychoeducation and was centered on behavioral techniques shown to be effective in the treatment of children with ASD and existing strategies used clinically in populations without ASD to address dental fear and maladaptive behaviors during dental care. An emphasis on optimizing parent engagement was embedded throughout treatment through the use of motivational interviewing to address expectancies, goal setting, and treatment barriers[56]; application of cognitive-behavioral techniques to enhance parenting self-efficacy[57]; and incorporation of mindful awareness to support nonjudgment of challenges related to dental care. Supplemental supports included a study library of video models, social stories, and parent handouts. Doctoral-level clinical or counseling psychologists and board-certified behavior analysts delivered treatment. The PT manual included detailed instructions and verbatim scripts. Therapists participated in annual in-

TABLE 1 Curriculum Overview of PT to Improve Dental Care in Children With ASD

Week	Session	Length, min	Content
1	Overview, participation enhancement, and introduction to behavioral principles	90	Conduct adapted participation enhancement intervention[56] to address parental expectations, motivation, goal setting, and treatment barriers. Provide treatment overview, review antecedent-behavior-consequence model and functions of behavior, assess oral aversions, and introduce antecedent interventions (eg, daily toothbrushing schedule, use of behavioral momentum, arranging the environment, providing choices).
2	Introduction to toothbrushing	90	Introduce priming as an antecedent intervention (visual schedule, video models, social story). Present toothbrushing task analysis and introduce prompting. Integrate use of consequences (reinforcement, planned ignoring). Provide video models and develop home dental support plan.
3	Toothbrushing with child	90	Introduce use of distraction and noncontingent escape (taking breaks). Present video models. Engage parent in in vivo practice with child. Refine home dental support plan.
4	Home coaching	60	Practice in vivo toothbrushing with child. Update home dental support plan as needed.
5	Flossing and diet	60	Introduce and demonstrate flossing (video models). Provide psychoeducation regarding diet and other factors influencing oral health. Update home dental support plan as needed. Forecast preparation for dental visit. Provide related video models.
6	Preparing for the dental visit	60	Present task analysis of dental visit and introduce desensitization practice. Review antecedent interventions, distraction, and noncontingent escape as applied to the dental office. Address planned use of consequences. Develop dental visit support plan.
8	Dental visit coaching	60	Engage parent in in vivo practice with child at the dental office. Update home and dental visit support plans as needed.
10	Phone booster 1	15–20	Review home practice and implementation of interventions. Refine home and dental visit support plans as needed.
12	3-month dental office visit		
13	Phone booster 2	15–20	Review home practice and implementation of interventions. Refine home and dental visit support plans as needed.
16	Phone booster 3	15–20	Review home practice and implementation of interventions. Refine home and dental visit support plans as needed.
21	Phone booster 4	15–20	Review home practice and implementation of interventions. Refine home and dental visit support plans as needed.
24	6-month dental office visit		

To provide flexibility and maximize dosage for underserved families, 8% of core sessions were completed after the 3-month follow-up assessment. Therapists also had the discretion to substitute a home visit for 1 phone booster session during the maintenance phase.

person cross-site training and weekly cross-site case supervision to ensure fidelity and integrity of PT delivery. Therapists also provided ratings of intervention fidelity and parental adherence after each intervention session.

Baseline Measures

Demographics and Family Information

Parents reported on parent and child age, sex, race, ethnicity, home language(s), marital status and cohabitation, household income, and child comorbid diagnoses, medication status, and intervention history.

Dental Experiences

Parents reported on children's current oral hygiene practices and parent and child previous dental visit history.

Intellectual Functioning

Child intellectual functioning was estimated using the Stanford-Binet 5 abbreviated battery IQ,[58] which yields a standard score with a mean of 100 and an SD of 15.

Adaptive Behavior

Adaptive behavior was measured using the Adaptive Behavior Composite score from the Vineland Adaptive Behavior Scales 3rd Edition Comprehensive Interview Form,[59] which has a mean of 100 and an SD of 15.

Overall Behavior Problems

Parent-reported overall child behavior problems were assessed using the Total Problems *T*-score from the Child Behavior Checklist (CBCL),[60] with higher scores denoting more behavior problems.

Primary Outcome Measures at Baseline, 3 Months, and 6 Months

Frequency of Home Toothbrushing

Parents reported the number of days in the past week that children completed toothbrushing twice per day, with or without help.

Visible Plaque

Dentists used a standard Visual Plaque Index (VPI)[55,61,62] to rate the buccal and lingual nonrestored surfaces of index teeth[63] on a 0 to 5 scale (0 = no plaque, 5 = plaque on more than two-thirds of tooth surface). Interrater reliability was adequate to excellent (82%–100% within 1 scale point).

Secondary Outcome Measures at Baseline, 3 Months, and 6 Months

Severity of Behavior Problems During Home Oral Hygiene

Parents reported on the occurrence and severity of 8 behavior problems during the past week's oral hygiene activities (not listening, difficulty sitting/standing still, actively resisting, eloping, fearful/anxious behaviors, screaming/yelling, aggression, and self-injury) using a 0 to 9 scale (0 = no problem, 9 = severe problem). Items were averaged to produce a single score ($\alpha = 0.91$).

Caries

Dentists completed the Decayed, Missing, and Filled Teeth Index (dmft/DMFT)[64] to document the presence and progression of full-mouth caries. The dmft/DMFT indicates the number of primary/permanent teeth that are decayed (d/D), missing due to caries (m/M), and filled (f/F). The d2/D2 code represents clinically detectable cavitated lesions. Interrater reliability was adequate to excellent ($\kappa = 0.61$–0.92).

Statistical Methods

Assuming at least 30% correlation between baseline and follow-up estimates and 2-tailed testing at $\alpha = 0.025$ given 2 primary outcomes, a minimum sample size of 100 was selected to provide 80% power to detect an effect size of 0.6. Based on trial variance estimates, this effect size corresponds to a 6-month treatment difference of 1.2 days/week in twice-daily toothbrushing and a 0.3-unit change in VPI. Enrollment targets permitted balanced attrition of 15%.

Analyses were intention to treat. Missing baseline data were mean imputed for 1 participant who exited the study before completing outcome measures. Treatment-dependent changes in outcomes were analyzed using shared baseline generalized mixed models with fixed effects for visit and child age, interaction terms for child age by visit and treatment by postbaseline visit, and random intercepts for each participant and site. Sensitivity analyses included main effects and interactions with visit for primary caregiver sex and baseline toothbrushing frequency. A Gaussian distribution with an identity link was used for continuous measures. A Poisson distribution with log link was used for dmft/DMFT and d2/D2 counts. A binomial distribution with a logit link was used for number of days of completed toothbrushing. Within-group changes from baseline and between-group differences were estimated using linear contrasts of least-squares means. Post hoc causal mediation analysis tested mediation of significant 3-month changes in oral health by concurrent changes in oral hygiene (twice-daily toothbrushing, electric toothbrush use, and severity of interfering behaviors). Percentages of total effect attributed to mediation are reported. Tests were 2-sided at $P < .025$ for primary outcomes and $P < .05$ for other comparisons. Analyses were performed using SAS version 9.4 software (SAS Institute Inc, Cary, NC).

Data Sharing

Individual deidentified participant data for the variables used in the analyses will be shared. The study protocol and statistical analysis plan will also be made available. Data will become available 18 months after the final study closeout date of August 31, 2020. Individuals interested in accessing the data should contact the corresponding author. Data will be made available to investigators pursuant Autism Intervention Network on Physical Health policies and institutional review board requirements.

RESULTS

Baseline Characteristics and Dental History

Primary caregivers tended to be female (90%) and married (53%), with a substantial proportion identifying as Hispanic/Latino (40%). The majority of children met *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition*, criteria for comorbid ID on the basis of study assessments (62%) and presented with clinically elevated behavior problems at baseline (61% CBCL *T*-score >63). Baseline characteristics were similar between randomly assigned groups, except PT primary caregivers were more often women, and PT families reported less frequent baseline oral hygiene than toolkit families (Table 2).

At baseline, 64% of families reported attempting toothbrushing daily, 51% reported completing brushing once per day, and 22% reported completing twice-daily toothbrushing. Most families used a manual toothbrush (75%), and 69% used fluoride toothpaste. A majority (75%) reported challenges with child behavior during toothbrushing. Regarding dental office care, 50% of children had received a past-year dental visit; use of pharmacologic interventions and physical restraint during previous visits was reported for 43% and 56%, respectively. For the 25 children (21%) without a previous dental visit, parental concerns about child behavior at the dental office emerged as the most commonly reported barrier (76%).

Intervention Attendance, Fidelity, and Adverse Events

Of the 58 families who received PT, 48 (83%) completed all core sessions 1 to 7, and 37 (64%) completed all booster sessions 8 to 11. Therapist-rated intervention fidelity (95%) and adherence (94%) were excellent across all PT sessions (1–11). No adverse events were identified.

TABLE 2 Baseline Demographics and Clinical Characteristics by Treatment Group

Variable	PT (n = 60)	Toolkit (n = 59)
Child		
Boys, n (%)	49 (82)	52 (88)
Age, y		
Mean (SD)	7.3 (2.6)	7.4 (2.7)
Range	3.1–13.0	3.0–13.3
Race, n (%)		
White	34 (61)	31 (61)
Asian	7 (13)	5 (10)
Black/African American	5 (9)	5 (10)
Hawaiian/Pacific Islander	2 (4)	1 (2)
American Indian/Alaskan Native	0	1 (2)
Multiracial	8 (14)	8 (16)
Ethnicity, n (%)		
Hispanic/Latino	22 (39)	20 (36)
IQ		
Mean (SD)	66.2 (19.5)	71.2 (20.3)
Range	47–112	47–124
Adaptive behavior, mean (SD)	57.1 (16.1)	57.8 (19.1)
ID, n (%)	41 (68)	33 (56)
ASD symptom severity, mean (SD)	7.3 (1.7)	7.2 (1.9)
Behavior problems		
CBCL T-score, mean (SD)	65 (9)	67 (10)
Clinically elevated (T-score >63), n (%)	32 (55)	38 (67)
Primary caregiver		
Age, y, mean (SD)	37 (7)	37 (9)
Women, n (%)	58 (97)	48 (83)*
Race, n (%)		
White	37 (74)	34 (68)
Asian	7 (14)	6 (12)
Black/African American	3 (6)	4 (8)
Hawaiian/Pacific Islander	1 (2)	0
American Indian/Alaskan Native	0	1 (2)
Multiracial	2 (4)	5 (10)
Ethnicity, n (%)		
Hispanic/Latino	24 (41)	22 (39)
Education, n (%)		
Less than eighth grade	1 (2)	3 (5)
Some high school	6 (10)	3 (5)
High school graduate	8 (14)	12 (21)
Some college	32 (54)	24 (42)
College degree	7 (12)	8 (14)
Advanced degree	5 (8)	7 (12)
Living situation, n (%)		
Married/domestic partner	32 (53)	30 (53)
Single/other adults in home	14 (23)	15 (26)
Single/no other adults in home	14 (23)	12 (21)
Annual gross family income, \$, n (%)		
<15 000	9 (16.1)	8 (15)
15 000–24 999	14 (25)	13 (24)
25 000–34 999	11 (20)	14 (26)
35 000–49 999	13 (23)	7 (13)
50 000–74 999	7 (13)	6 (11)
≥75 000	2 (4)	6 (11)
Primary home language English, n (%)	50 (85)	47 (89)
Baseline home oral hygiene		
Past-week twice-daily toothbrushing, mean (SD)	2.48 (2.51)	3.74 (2.90)*
Behavior problem severity during oral hygiene, mean (SD)	2.92 (2.39)	2.78 (2.33)
Baseline oral health		
Visible plaque, mean (SD)	0.95 (0.68)	0.77 (0.50)
Caries, mean (SD)	3.19 (4.55)	2.72 (3.21)
No. of decayed teeth, mean (SD)	1.19 (2.12)	0.67 (1.39)
Child age at first dental visit, y, mean (SD)	3 (2)	3 (2)

*P < .05.

Primary Outcomes

Results for primary and secondary outcomes are presented in Table 3 and Fig 2. Frequency of twice-daily toothbrushing improved significantly within each group. However, gains were significantly greater at both time points for children in PT relative to the toolkit intervention. Only PT significantly decreased visible plaque. PT resulted in greater plaque reduction than the toolkit intervention at 3 months. Group differences fell below significance at 6 months.

Secondary Outcomes

In the PT group, parents reported significantly reduced severity of child behavior problems during home oral hygiene activities at both time points. In the toolkit group, parents reported reduced behavior problems at 6 months only. Significant reductions in child behavior problems in the PT relative to the toolkit intervention were reported at both 3 and 6 months.

Children in toolkit group had significantly more caries over time than children in the PT group. At 3 months, group differences in full-mouth caries were significant. Although the toolkit intervention continued to exhibit a greater relative increase in full-mouth caries and decayed teeth over time, differences fell short of significance at 6 months.

Sensitivity Analyses

Adjustment for baseline differences revealed an identical pattern of results, with a slight reduction in estimated group differences in 3-month caries (Supplemental Table 4).

Oral Hygiene as a Mediator

Neither improved frequency of oral hygiene (VPI, −2% [95% confidence interval [CI], −25% to 21%; P = .86]; dmft/DMFT, −8% [95% CI, −30% to 15%; P = .51]), increased use of an electric toothbrush (VPI, 0.2% [95% CI, −5% to 5%; P = .95]; dmft/DMFT, 0.5%, 95% CI, −9% to 8%; P = .92), nor reduction of behavior problems during oral hygiene (VPI, 22% [95% CI, −19% to 63%; P = .29]; dmft/DMFT, 21% [95% CI, −20% to 61%; P = .32]) significantly mediated 3-month PT effects on plaque and caries. However, the 95% CI for behavior problems included mediation effects that would be considered substantial.

DISCUSSION

Both PT and a psychoeducational dental toolkit were associated with improved frequency of toothbrushing, but PT was associated with greater relative gains in oral hygiene and a consistent reduction in the severity of interfering child behavior problems. Moreover, PT was associated with improved oral health by reducing plaque and minimizing progression of caries at immediate follow-up, whereas plaque was not reduced, and caries significantly worsened with the toolkit intervention. To our

TABLE 3 Oral Hygiene and Oral Health Outcomes at 3 and 6 Months

	PT (n = 60)		Toolkit (n = 59)		
Outcome and Time Point	Estimate (95% CI)	Effect (95% CI)	Estimate (95% CI)	Effect (95% CI)	Group Difference Intervention Effect (95% CI)
Home oral hygiene					
Frequency of twice-daily tooth brushing[a]					
Baseline		—		—	—
Days	2.8 (2.2 to 3.5)	—	2.8 (2.2 to 3.5)	—	—
%	40 (31 to 50)	—	40 (31 to 50)	—	—
3 mo	—	5.16*** (3.59 to 7.40)	—	1.83** (1.27 to 2.63)	2.82*** (1.71 to 4.64)
Days	5.4 (4.8 to 5.9)	—	3.9 (3.1 to 4.7)	—	—
%	78 (69 to 85)	—	55 (44 to 67)	—	—
6 mo	—	5.26*** (3.63 to 7.62)	—	2.39*** (1.66 to 3.44)	2.20** (1.33 to 3.65)
Days	5.5 (4.8 to 6.0)	—	4.3 (3.5 to 5.1)	—	—
%	78 (69 to 85)	—	62 (50 to 72)	—	—
Severity of behavior problems during oral hygiene activities					
Baseline	2.89 (2.51 to 3.27)	—	2.89 (2.51 to 3.27)	—	—
3 mo	1.71 (1.21 to 2.22)	−1.18*** (−1.64 to −0.71)	2.62 (2.12 to 3.11)	−0.28 (−0.73 to 0.18)	−0.90** (−1.52 to −0.28)
6 mo	1.37 (0.86 to 1.88)	−1.52*** (−1.99 to −1.05)	2.14 (1.64 to 2.64)	−0.75** (−1.21 to −0.29)	−0.77* (−1.39 to −0.14)
Oral health status					
Visible plaque					
Baseline	0.84 (0.75 to 0.94)	—	0.84 (0.75 to 0.94)	—	—
3 mo	0.68 (0.55 to 0.81)	−0.17* (−0.29 to −0.04)	0.87 (0.74 to 1.00)	0.03 (−0.10 to 0.15)	−0.19* (−0.36 to −0.02)
6 mo	0.75 (0.62 to 0.88)	−0.09 (−0.22 to 0.04)	0.80 (0.67 to 0.92)	−0.05 (−0.17 to 0.08)	−0.05 (−0.21 to 0.12)
Full-mouth caries (dmft/DMFT)[b]					
Baseline	0.88 (0.33 to 2.33)	—	0.88 (0.33 to 2.33)	—	—
3 mo	0.84 (0.31 to 2.26)	0.95 (0.75 to 1.21)	1.14 (0.43 to 3.06)	1.30* (1.03 to 1.63)	0.73* (0.54 to 0.99)
6 mo	0.93 (0.35 to 2.50)	1.06 (0.83 to 1.34)	1.09 (0.41 to 2.91)	1.23† (0.98 to 1.55)	0.86 (0.63 to 1.17)
No. of decayed teeth (d2/D2)[b]					
Baseline	0.29 (0.19 to 0.45)	—	0.29 (0.19 to 0.45)	—	—
3 mo	0.34 (0.21 to 0.55)	1.16 (0.82 to 1.66)	0.47 (0.29 to 0.77)	1.63* (1.11 to 2.39)	0.71 (0.43 to 1.19)
6 mo	0.34 (0.20 to 0.55)	1.16 (0.81 to 1.66)	0.51 (0.31 to 0.83)	1.75** (1.18 to 2.58)	0.66 (0.40 to 1.11)

—, not applicable.

†$P < .10$; *$P < .05$; **$P < .01$; ***$P < .001$.

[a] Binomial model with logit link presents estimates as n (%) of days per week and effects as ORs or intervention effects as ratio of ORs.

[b] Poisson model with log link presents estimates as counts and effects as rate ratios or intervention effects as ratio of rate ratios.

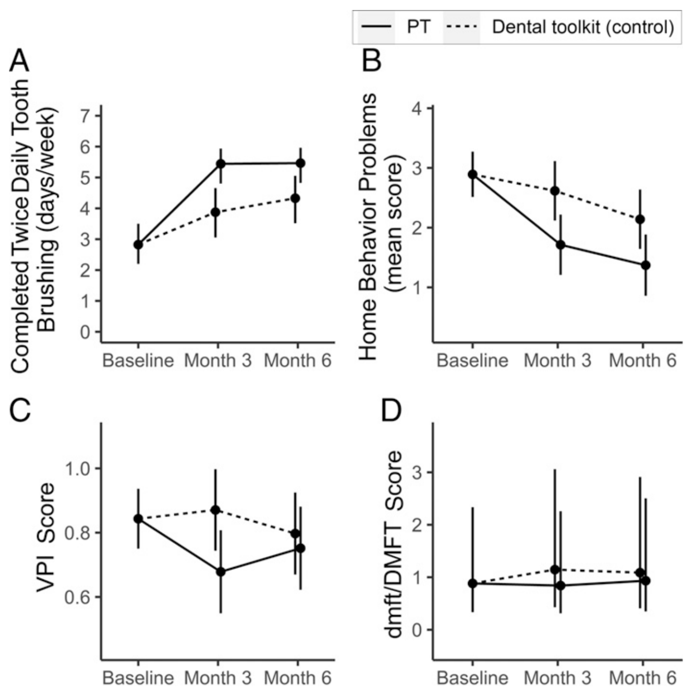

FIGURE 2
Estimated means with 95% CIs for oral hygiene and oral health outcomes.

knowledge, this RCT is the first to reveal significant treatment effects on oral hygiene and oral health in children with ASD. Beneficial effects of PT suggest that this approach is viable for addressing one of the most common unmet health care needs of children with ASD: problematic oral hygiene and poor oral health.[2–7]

The PT intervention was numerically superior to the toolkit intervention for all outcomes assessed, which represents a stringent test of PT advantage given the multiple interventions contained in our active comparator. The toolkit was associated with meaningful improvement in frequency of oral hygiene and some reduction of child behavior problems during home dental care. Indeed, toolkit improvements may partly account for the reduced group differences in oral health seen at 6 months. Evidence of meaningful benefits from the relatively low-touch toolkit highlights the need to provide underserved families with access to high-quality, user-friendly psychoeducational materials and basic oral hygiene resources.

One-half of our sample had not visited the dentist in the past year, and 21% had never been to the dentist. For these families in particular, study facilitation of 3 dental visits may have functioned as an additional intervention. Our study did not focus on barriers to preventive dental visits. However, delivery of appropriate and timely access to dental services remains a critical need for children with ASD, which may be remedied in part through adoption of comprehensive care coordination.[17]

Poor oral hygiene is posited to underlie oral health problems in children with ASD.[6,8] Although mediation results were nonsignificant, CIs included potentially substantial effects for child behavior problems, suggesting the possibility that improved behavior management may reflect a core treatment mechanism. Future trials would benefit from being powered specifically for mediation testing and from using measures with enhanced sensitivity to detect change in quality of oral hygiene. Integrating smart toothbrush technology may assist with measurement precision.

Strengths of the study include our use of a multisite RCT design with an active comparator, blinded dentist evaluation of oral health, and follow-up at 3 and 6 months after baseline. We also demonstrated strong retention of our underserved sample comprising families with Medicaid eligibility. The socioeconomic and ethnic diversity of our sample is rare in RCTs for children with ASD conducted in the United States.[65,66] Moreover, the majority of the participating children with ASD presented with comorbid ID, a subpopulation that is strikingly underrepresented in research literature.[67]

Consistent with existing studies, we relied on unblinded parent report of toothbrushing frequency. Use of an active comparator minimized potential response bias, but future work would benefit from multimethod measurement and use of a comparator matched for contact and intensity. Previously significant oral health differences also lessened at follow-up, which may suggest the need for greater PT supports during maintenance. Alternatively, long-term follow-up might yield more pronounced treatment effects because caries progression slowed in the PT group and increased in the toolkit group. PT oral health improvements, although statistically significant, were relatively small, and it is possible that augmenting oral health measures would enable detection of larger and more clinically significant effects. To reduce participant burden, dentists used plaque ratings of index teeth to represent full dentition.[63] An examination of plaque on all teeth might enhance sensitivity. Finally, the results reveal the efficacy of PT as administered by highly trained therapists with a select population. Community-based replication is needed.

CONCLUSIONS

The PT intervention was efficacious relative to a psychoeducational dental toolkit for improving oral hygiene and oral health in underserved children with ASD. The PT intervention also reduced the severity of child behavior problems interfering with home oral hygiene. Given the feasibility and preliminary efficacy of our manualized PT program, we recommend additional study of our novel application of PT to promote oral health in children with ASD, a population at high risk for unmet dental needs.

ACKNOWLEDGMENTS

We thank the members of the clinical coordinating center and data coordinating center at Massachusetts General Hospital, including Lily Altstein, Justin Farmer, Siobhan Glynn, Hilda Gutierrez, Richard Morse, James Perrin, and Brian Winklosky, as well as the members of the Massachusetts General Hospital data safety and monitoring board, including Ann Neumeyer (chair), Amanda Bennett, and Larry Lin. Additionally, we thank project team members at each site, including Charles Astley, Melissa Brown, Breana Cervantes, Jean Gehricke, Taylor Harlan, Haley Kitchens, Anna Laakman, Michele Ocen, Meghan Orr, Erin Roby, and Lisa Tully, and gratefully acknowledge the contributions of Cathy Flaitz, Tanya Mathew, David McAdam, Richard Udin, Konita Wilks, and all the families who made this work possible. This work was conducted through the Autism Speaks Autism Treatment Network serving as the Autism Intervention Research Network on Physical Health.

ABBREVIATIONS

ASD: autism spectrum disorder
CBCL: Child Behavior Checklist
CI: confidence interval
dmft/DMFT: Decayed, Missing, and Filled Teeth Index
ID: intellectual disability
PT: parent training
RCT: randomized controlled trial
VPI: Visual Plaque Index

PEDIATRICS (ISSN Numbers: Print, 0031-4005; Online, 1098-4275).

conceptualized and designed the study and critically reviewed and revised the manuscript; Drs Albright, Guijon, Hammersmith, Scherr, Spaulding, and Stephenson and Ms Moffitt supported development of the data collection instruments, collected data, and critically reviewed and revised the manuscript; Ms McKinnon-Bermingham developed the treatment curriculum, collected data, and critically reviewed and revised the manuscript; Mr Chan performed analyses, drafted the initial manuscript, and critically reviewed and revised the manuscript; Ms Lu supervised data collection and critically reviewed and revised the manuscript; and all authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

This trial has been registered at www.clinicaltrials.gov (identifier NCT03003221).

A randomized controlled trial revealed efficacy of a novel parent training intervention for improving oral hygiene and oral health in underserved children with autism spectrum disorder.

DOI: https://doi.org/10.1542/peds.2021-050691

Accepted for publication Feb 1, 2022

Address correspondence to Rachel M. Fenning, PhD, Department of Child and Adolescent Studies and Center for Autism, California State University, Fullerton, 800 N State College Blvd, EC 560, Fullerton, CA 92831. E-mail: rfenning@fullerton.edu

PEDIATRICS (ISSN Numbers: Print, 0031-4005; Online, 1098-4275).

FUNDING: This project was supported by the Health Resources and Services Administration of the US Department of Health and Human Services under cooperative agreement UA3 MC11054–Autism Intervention Research Network on Physical Health. The funding source had no role in the design or conduct of the study; data collection, management, analysis, or interpretation; and preparation, review, approval, or submission of the manuscript. This information or content and conclusions are those of the authors and should not be construed as the official position or policy of, nor should any endorsements be inferred by, the Health Resources and Services Administration, US Department of Health and Human Services, or the US government.

CONFLICT OF INTEREST DISCLOSURES: Dr Macklin reports serving on advisory boards for Biogen, Cerevance, and Stoparkinson Healthcare Systems, serving on data monitoring boards for Novartis and Takeda Pharmaceuticals, and receiving research funds through his institution on his behalf from Amylyx Pharmaceuticals, Biohaven Pharmaceuticals, Clene Nanomedicine, GlaxoSmithKline, Mitsubishi Tanabe Pharma America, Prilenia Therapeutics, Ra Pharmaceuticals, and the National Institutes of Health, and Dr Coury has received research grant support from GW Biosciences, Stalicla, and Stemina and serves on advisory boards for BioRosa, Cognoa, GW Biosciences, MaraBio, Quadrant, and Stalicla; the other authors have indicated they have no conflicts of interest to disclose.

REFERENCES

1. McKinney CM, Nelson T, Scott JM, Heaton LJ, Vaughn MG, Lewis CW. Predictors of unmet dental need in children with autism spectrum disorder: results from a national sample. *Acad Pediatr.* 2014;14(6):624–631
2. Bhandary S, Hari N. Salivary biomarker levels and oral health status of children with autistic spectrum disorders: a comparative study. *Eur Arch Paediatr Dent.* 2017;18(2):91–96
3. DeMattei R, Cuvo A, Maurizio S. Oral Assessment of children with an autism spectrum disorder. *J Dent Hyg.* 2007;81(3):65
4. Jaber MA, Sayyab M, Abu Fanas SH. Oral health status and dental needs of autistic children and young adults. *J Investig Clin Dent.* 2011;2(1):57–62
5. Kopycka-Kedzierawski DT, Auinger P. Dental needs and status of autistic children: results from the National Survey of Children's Health. *Pediatr Dent.* 2008;30(1):54–58
6. Marshall J, Sheller B, Mancl L. Caries-risk assessment and caries status of children with autism. *Pediatr Dent.* 2010;32(1):69–75
7. Pi X, Liu C, Li Z, Guo H, Jiang H, Du M. A meta-analysis of oral health status of children with autism. *J Clin Pediatr Dent.* 2020;44(1):1–7
8. Lam PPY, Du R, Peng S, McGrath CPJ, Yiu CKY. Oral health status of children and adolescents with autism spectrum disorder: a systematic review of case-control studies and meta-analysis. *Autism.* 2020;24(5):1047–1066
9. Du RY, Yiu CKY, King NM, Wong VCN, McGrath CPJ. Oral health among preschool children with autism spectrum disorders: a case-control study. *Autism.* 2015;19(6):746–751
10. Du RY, Yiu CKY, King NM. Oral health behaviours of preschool children with autism spectrum disorders and their barriers to dental care. *J Autism Dev Disord.* 2019;49(2):453–459
11. Stein LI, Lane CJ, Williams ME, Dawson ME, Polido JC, Cermak SA. Physiological and behavioral stress and anxiety in children with autism spectrum disorders during routine oral care. *BioMed Res Int.* 2014;2014:694876
12. Stein LI, Polido JC, Najera SOL, Cermak SA. Oral care experiences and challenges in children with autism spectrum disorders. *Pediatr Dent.* 2012;34(5):387–391
13. Eidelman E, Faibis S, Peretz B. A comparison of restorations for children with early childhood caries treated under general anesthesia or conscious sedation. *Pediatr Dent.* 2000;22(1):33–37
14. Wilson S. Management of child patient behavior: quality of care, fear and anxiety, and the child patient. *J Endod.* 2013;39(3 suppl):S73–S77
15. Lai B, Milano M, Roberts MW, Hooper SR. Unmet dental needs and barriers to dental care among children with autism spectrum disorders. *J Autism Dev Disord.* 2012;42(7):1294–1303
16. Brickhouse TH, Farrington FH, Best AM, Ellsworth CW. Barriers to dental care for children in Virginia with autism spectrum disorders. *J Dent Child (Chic).* 2009;76(3):188–193
17. Fenning RM, Steinberg-Epstein R, Butter EM, et al. Access to dental visits and correlates of preventive dental care in children with autism spectrum disorder. *J Autism Dev Disord.* 2020;50(10):3739–3747
18. Cagetti MG, Mastroberardino S, Campus S, et al. Dental care protocol based on visual supports for children with autism spectrum disorders. *Med Oral Patol Oral Cir Bucal.* 2015;20(5):e598–e604
19. Cermak SA, Stein Duker LI, Williams ME, Dawson ME, Lane CJ, Polido JC. Sensory adapted dental environments to enhance oral care for children with autism spectrum disorders: a randomized controlled pilot study. *J Autism Dev Disord.* 2015;45(9):2876–2888
20. Isong IA, Rao SR, Holifield C, et al. Addressing dental fear in children with autism spectrum disorders: a randomized controlled pilot study using electronic screen media. *Clin Pediatr (Phila).* 2014;53(3):230–237
21. Orellana LM, Martínez-Sanchis S, Silvestre FJ. Training adults and children with an autism spectrum disorder to be compliant with a clinical dental assessment using a TEACCH-based approach. *J Autism Dev Disord.* 2014;44(4):776–785
22. Bäckman B, Pilebro C. Visual pedagogy in dentistry for children with autism. *ASDC J Dent Child.* 1999;66(5):325–331, 294
23. Hernandez P, Ikkanda Z. Applied behavior analysis: behavior management of children with autism spectrum disorders in dental environments. *J Am Dent Assoc.* 2011;142(3):281–287
24. Lefer G, Rouches A, Bourdon P, Lopez Cazaux S. Training children with autism spectrum disorder to undergo oral assessment using a digital iPad® application. *Eur Arch Paediatr Dent.* 2019;20(2):113–121
25. Luscre DM, Center DB. Procedures for reducing dental fear in children with autism. *J Autism Dev Disord.* 1996;26(5):547–556
26. Mah JWT, Tsang P. Visual schedule system in dental care for patients with autism: a pilot study. *J Clin Pediatr Dent.* 2016;40(5):393–399
27. Murshid EZ. Effectiveness of a preparatory aid in facilitating oral assessment in a group of Saudi children with autism spectrum disorders in Central Saudi Arabia. *Saudi Med J.* 2017;38(5):533–540
28. Nelson T, Chim A, Sheller BL, McKinney CM, Scott JM. Predicting successful dental examinations for children with autism spectrum disorder in the context of a dental desensitization program. *J Am Dent Assoc.* 2017;148(7):485–492
29. Nelson TM, Sheller B, Friedman CS, Bernier R. Educational and therapeutic behavioral approaches to providing dental care for patients with autism spectrum disorder. *Spec Care Dentist.* 2015;35(3):105–113
30. Nilchian F, Shakibaei F, Jarah ZT. Evaluation of visual pedagogy in dental check-ups and preventive practices among 6-12-year-old children with autism. *J Autism Dev Disord.* 2017;47(3):858–864
31. Orellana LM, Cantero-Fuentealba C, Schmidlin-Espinoza L, Luengo L. Psychoeducational intervention to improve oral assessment in people with autism

spectrum disorder, BIO-BIO region, Chile. *Med Oral Patol Oral Cir Bucal.* 2019;24(1):e37–e46

32. Zink AG, Diniz MB, Rodrigues Dos Santos MTB, Guaré RO. Use of a picture exchange communication system for preventive procedures in individuals with autism spectrum disorder: pilot study. *Spec Care Dentist.* 2016;36(5):254–259
33. Delli K, Reichart PA, Bornstein MM, Livas C. Management of children with autism spectrum disorder in the dental setting: concerns, behavioural approaches and recommendations. *Med Oral Patol Oral Cir Bucal.* 2013;18(6):e862–e868
34. Sarnat H, Samuel E, Ashkenazi-Alfasi N, Peretz B. Oral health characteristics of preschool children with autistic syndrome disorder. *J Clin Pediatr Dent.* 2016;40(1):21–25
35. Stein LI, Polido JC, Cermak SA. Oral care and sensory over-responsivity in children with autism spectrum disorders. *Pediatr Dent.* 2013;35(3):230–235
36. Agouropoulos A, Twetman S, Pandis N, Kavvadia K, Papagiannoulis L. Caries-preventive effectiveness of fluoride varnish as adjunct to oral health promotion and supervised tooth brushing in preschool children: a double-blind randomized controlled trial. *J Dent.* 2014;42(10):1277–1283
37. Chestnutt IG, Schäfer F, Jacobson AP, Stephen KW. The influence of toothbrushing frequency and post-brushing rinsing on caries experience in a caries clinical trial. *Community Dent Oral Epidemiol.* 1998;26(6):406–411
38. Hilgert LA, Leal SC, Bronkhorst EM, Frencken JE. Long-term effect of supervised toothbrushing on levels of plaque and gingival bleeding among schoolchildren. *Oral Health Prev Dent.* 2017;15(6):537–542
39. Adair PM, Pine CM, Burnside G, et al. Familial and cultural perceptions and beliefs of oral hygiene and dietary practices among ethnically and socio-economically diverse groups. *Community Dent Health.* 2004;21(1 suppl):102–111
40. Poutanen R, Lahti S, Tolvanen M, Hausen H. Parental influence on children's oral health-related behavior. *Acta Odontol Scand.* 2006;64(5):286–292
41. Saied-Moallemi Z, Virtanen JI, Ghofranipour F, Murtomaa H. Influence of mothers' oral health knowledge and attitudes on their children's dental health. *Eur Arch Paediatr Dent.* 2008;9(2):79–83
42. Bal VH, Kim S-H, Cheong D, Lord C. Daily living skills in individuals with autism spectrum disorder from 2 to 21 years of age. *Autism.* 2015;19(7):774–784
43. Eaves LC, Ho HH. Young adult outcome of autism spectrum disorders. *J Autism Dev Disord.* 2008;38(4):739–747
44. Matson JL, Hattier MA, Belva B. Treating adaptive living skills of persons with autism using applied behavior analysis: a review. *Res Autism Spectr Disord.* 2012;6(1):271–276
45. Bishop MR, Kenzer AL, Coffman CM, Tarbox CM, Tarbox J, Lanagan TM. Using stimulus fading without escape extinction to increase compliance with toothbrushing in children with autism. *Res Autism Spectr Disord.* 2013;7(6):680–686
46. Lopez Cazaux S, Lefer G, Rouches A, Bourdon P. Toothbrushing training programme using an iPad® for children and adolescents with autism. *Eur Arch Paediatr Dent.* 2019;20(3):277–284
47. Sallam A, Badr S, Rashed M. Effectiveness of audiovisual modeling on the behavioral change toward oral and dental care in children with autism. *Indian J Dent.* 2013;4(4):184–190
48. Zhou N, Wong HM, McGrath C. Efficacy of social story intervention in training toothbrushing skills among special-care children with and without autism. *Autism Res.* 2020;13(4):666–674
49. Al-Batayneh OB, Nazer TS, Khader YS, Owais AI. Effectiveness of a tooth-brushing programme using the picture exchange communication system (PECS) on gingival health of children with autism spectrum disorders. *Eur Arch Paediatr Dent.* 2020;21(2):277–283
50. Pilebro C, Bäckman B. Teaching oral hygiene to children with autism. *Int J Paediatr Dent.* 2005;15(1):1–9
51. Popple B, Wall C, Flink L, et al. Brief report: remotely delivered video modeling for improving oral hygiene in children with ASD: a pilot study. *J Autism Dev Disord.* 2016;46(8):2791–2796
52. Dias GG, Prado EFG, Vadasz E, Siqueira JTT. Evaluation of the efficacy of a dental plaque control program in autistic patients. *J Autism Dev Disord.* 2010; 40(6):704–708
53. American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*, 5th ed. Arlington, VA: American Psychiatric Association; 2013
54. Lord C, Rutter M. *Autism Diagnostic Observation Schedule*, 2nd ed. Torrance, CA: Western Psychological Services; 2012
55. Yaacob M, Worthington HV, Deacon SA, et al. Powered versus manual toothbrushing for oral health. *Cochrane Database Syst Rev.* 2014;(6):CD002281
56. Nock MK, Kazdin AE. Randomized controlled trial of a brief intervention for increasing participation in parent management training. *J Consult Clin Psychol.* 2005;73(5):872–879
57. Durand V, Hieneman M. *Helping Parents With Challenging Children: Positive Family Intervention.* New York, NY: Oxford University Press; 2008
58. Roid G. *Stanford-Binet Intelligence Scales*, 5th ed. Itasca, IL: Riverside Publishing; 2003
59. Sparrow S, Cicchetti D, Saulnier C. *Vineland Adaptive Behavior Scales*, 3rd ed. Bloomington, MN: NCS Pearson; 2016
60. Achenbach T. *The Achenbach System of Empirically Based Assessment (ASEBA): Development, Findings, Theory, and Applications.* Burlington, VT, ASEBA; 2009
61. Quigley GA, Hein JW. Comparative cleansing efficiency of manual and power brushing. *J Am Dent Assoc.* 1962;65:26–29
62. Turesky S, Gilmore ND, Glickman I. Reduced plaque formation by the chloromethyl analogue of victamine C. *J Periodontol.* 1970;41(1):41–43
63. Gettinger G, Patters MR, Testa MA, et al. The use of six selected teeth in population measures of periodontal status. *J Periodontol.* 1983;54(3):155–159

64. World Health Organization (WHO). *Oral Health Surveys: Basic Methods*, 5th ed. Geneva, Switzerland; World Health Organization; 2013:47
65. Kasari C, Lawton K, Shih W, et al. Caregiver-mediated intervention for low-resourced preschoolers with autism: an RCT. *Pediatrics*. 2014;134(1):e72–e79
66. Safer-Lichtenstein J, Hamilton JC, McIntyre LL. Examining demographics in randomized controlled trials of group-based social skills interventions for individuals with autism spectrum disorder. *J Autism Dev Disord*. 2019;49(8):3453–3461
67. Russell G, Mandy W, Elliott D, White R, Pittwood T, Ford T. Selection bias on intellectual ability in autism research: a cross-sectional review and meta-analysis. *Mol Autism*. 2019;10(1):9